

# ORGANIC FARMING

*Is submitted to*

**Jawaharlal Nehru Technological University Anantapur,  
Ananthapuramu.**

*In partial fulfilment of  
the requirements for the award of the degree of*

**BACHELOR OF TECHNOLOGY**

**In**

**COMPUTER SCIENCE AND ENGINEERING**

**Submitted By**

**NAME OF THE STUDENT**

**ROLL NO**

**SHAIK MOHAMMAD RAYYAN**

**238X1A05A9**

*Under the esteemed guidance of*

Mr. K. Hari, M.Tech ,

Asst Professor



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**MJR COLLEGE OF ENGINEERING AND TECHNOLOGY**

**(Affiliated to JNTUA, Ananthapuramu and Approved by AICTE, New Delhi)**

**Diguvapokulavaripalli, Pulicherla (Mandal), Chittoor (D) -517 113, A.P**

**2023-2027**

# **MJR COLLEGE OF ENGINEERING AND TECHNOLOGY**

(Approved by AICTE, New Delhi and Affiliated to JNTU, Anantapur)

Diguvapokulavari Palli, Pulicherla(M), Chittoor(Dist.), Andhra

Pradesh-517113.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

## **BONAFIDE CERTIFICATE**

This is to certify that the Community Services project entitled “**ORGANIC FARMING**” is a bonafide work done by **SHAIK MOHAMMAD RAYYAN** registerno.**238X1A05A9**. In partial fulfillment for the degree in B.Tech (Computer Science) to the Jawaharlal Nehru Technological University Anantapur during the Academic year 2023–2027

### **GUIDE**

MR. K. HARI

(M.Tech. Asst. Professor)

### **HEAD OF THE DEPARTMENT**

Dr. K.G. CHIRANJEEVI

(M.Tech, Ph.d), Professor

**Program Book**  
**For**  
**Community Service Project**  
**On**  
**ORGANIC FARMING**

**Name of the student:**

**SHAIK MOHAMMAD RAYYAN**

**Roll number:**

**238X1A05A9**

Name of the College: **MJR COLLEGE OF ENGINEERING AND TECHNOLOGY,**

(Approved by A.I.C.T.E., New Delhi and Affiliated to J.N.T.U.A,

Anantapur), Diguva pokulavari Palli, Pulicherla (M), Chittoor district-

517113)

Period of CSP: From :05-05-2025 To : 05-07-2025

Name & Address of the Community/Habitation: **PILERU**

# Community Service Project Report

Submitted in accordance with the requirement for the degree Of B.Tech

Name of the College: MJR COLLEGE OF ENGINEERING AND TECHNOLOGY

Department: COMPUTER SCIENCE AND ENGINEERING

Name of the Faculty Guide: Mr. K. Hari

Duration of the CSP: From: 05-05-2025 To 05-07-2025

Year of Study: II Year Of B.Tech

**Name of the student**

SHAIK MOHAMMAD RAYYAN

**Rollnumber**

238X1A05A9

## **Student's Declaration**

We are, a student of **II B.Tech** Program, Reg. No 238X1A05A9 of the department of **COMPUTER SCIENCE AND ENGINEERING**, from MJRA do here by declare that I have completed the mandatory community service from 05-05-2025 to 05-07-2025 in piler (name of the community / habitation) under the faculty guide ship of Mr. K. Hari, department of Computer Science and Engineering in college MJR COLLEGE OF ENGINEERING AND TECHNOLOGY

(Signature and Date)

Faculty Guide

Head of the Department

Principal

## Certificate from Official of the Community

This is to certify that **SHAIK MOHAMMAD RAYYAN** Reg.No:**238X1A05A9**, of **MJR COLLEGE OF ENGINEERING AND TECHNOLOGY** underwent community service in PILERU (mandal) farmers community (PILER) From:05- 05-2025 To: 05- 07- 2025

The overall performance of the Community Service Volunteer during his/her community service is found to be (Satisfactory/Good).

Authorized Signatory with Date and Seal

## ACKNOWLEDGEMENTS

I wish to express my sincere thanks to our chairman sir **M. P. AVINASH KUMAR REDDY piler, chittoor Dist**, our beloved vice chairman for having provided all the facilities and support in completing my project successfully.

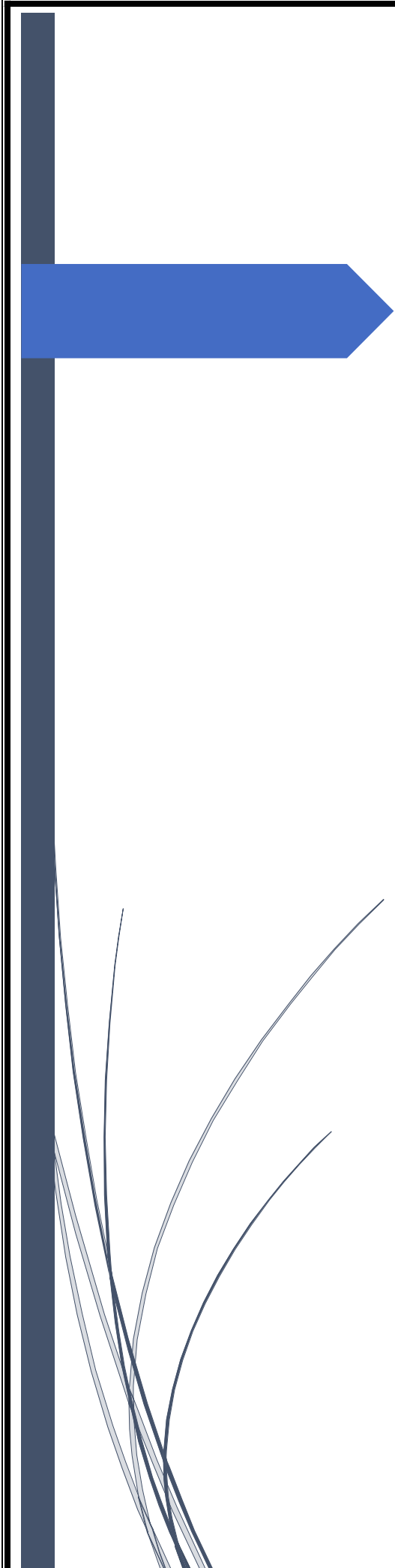
The completion of project brings with and sense of satisfaction, but it is never completed without thinking the persons who are all responsible for its successful completion. I wish to express my profound feelings of graduate to this grade instructions of ours **MJRA COLLEGE OF ENGINEERING AND TECHNOLOGY** for providing us the excellent faculties.

I earnestly thank **DR. K V N V N Rao sir**, principal for having permitted me to carry out the project on Herbal Plants successfully.

I extremely thankful to **DR. K. G. Chiranjeevi sir., Ph. D., professor and Head of the Department of Computer Science and Engineering**, for the valuable support given to me, without whom we would not have been able to complete my project work.

I wish to express my deep sense of gratitude to internal guide to **Mr. K Hari., M.Tech., Associate Professor**, for her valuable guidance and useful suggestion at all stages of the project, which helped me in completing the project work in time.

(SHAIK MOHAMMAD RAYYAN  
RegNo:238X1A05A9)



# **PROGRAM BOOK**

**Community Service Project**  
**on**  
**ORGANIC FARMING**

**AP STATE COUNCIL OF HIGHER EDUCATION**  
**A STATUTORY BODY OF GOVERNMENT OF ANDHRA PRADESH**



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# CHAPTER 1: EXECUTIVE SUMMARY

As part of the Community Service Project (CSP) under the curriculum of MJR College of Engineering and Technology, our team of five second-year B.Tech CSE students undertook a project on Organic Farming. The aim was to understand and promote sustainable agricultural practices in a rural community setting.

We engaged with the local farming community in and around PILERU where tradition and organic farming methods are being adopted. Through field visits, interactive sessions with farmers, and hands-on involvement, we gained the insights into natural composting, pest control using organic methods, crop rotation, and water management practices.

## Activities Performed:

- Visited local organic farms to study traditional methods.
- Participated in compost preparation using cow dung and vegetable waste.
- Assisted farmers in seed sowing organic pest control application.
- Conducted an awareness session on organic farming benefits with local youth.
- Documented best practices and shared insights with fellow students.

## Learning Objectives and Outcomes:

### 1. Understanding

Sustainable Agriculture: Gained knowledge about eco-friendly farming techniques and their benefits to health and the environment.

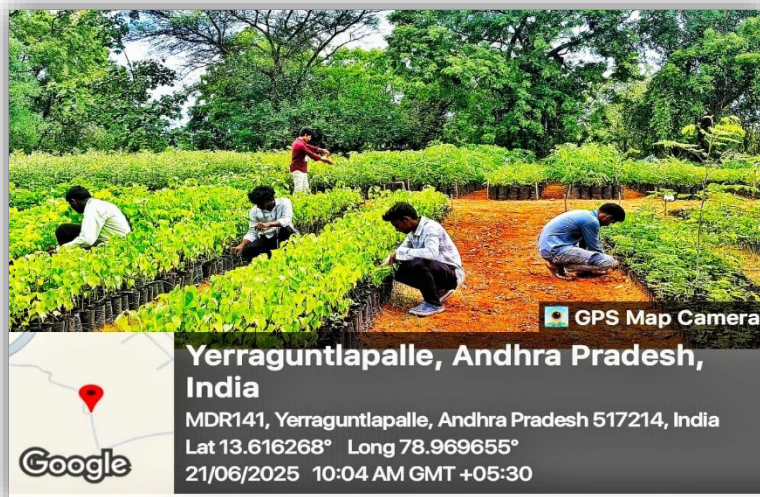
2. Community Engagement Skills: Developed communication and collaboration skills by interacting with farmers and community leaders.

3. Teamwork and Leadership: Worked effectively as a team, managed responsibilities, and practiced leadership in real-life scenarios.

4. Problem Solving in Real Environments: Identified practical issues faced by farmers and explored low-cost organic solutions.

5. Environmental Awareness: Realized the importance of protecting soil health, reducing chemical use, and promoting biodiversity.

This project not only enriched our academic learning but also instilled a sense of social lost of responsibility and the environmental consciousness. It was a valuable experience that bridged theoretical knowledge with practical understanding.



**Fig:** Visited local organic farms to study.

## CHAPTER 2: OVERVIEW OF THE COMMUNITY

### About the community- PILERU, Annamayya District

Our Community Service Project was carried out in PILERU, a town located in Anantapur district, Andhra Pradesh. Pileru is a semi-urban area surrounded by several rural habitations and agricultural lands. The name "Pileru" is believed to have originated from the local river Pillairu, which supports irrigation in the region. Historically, pileru has served as a regional hub for agriculture, trade, and education. It is known for its production of crops such as paddy, groundnut, and vegetables. In recent years, a growing interest in organic and natural farming has emerged among some progressive farmers in the area.

The community in Pileru is culturally diverse, with people from various social and economic backgrounds coexisting peacefully. The traditions and festivals—particularly Sankranti, Ugadi, and the Diwali—are celebrated with great enthusiasm, often centered around agricultural themes. The people are deeply rooted in values such as respect for elders, hospitality, communal harmony, and a strong connection to the land. Ethically, the community upholds a spirit of cooperation and simplicity, which made our engagement with them smooth and rewarding. Their interest in sustainable practices like organic farming reflects an alignment between tradition and ecological responsibility.

### Socio-Economic Conditions of the Community

The socio-economic profile of Pileru reflects a mix of both rural and urban lifestyles. Agriculture remains the primary source of income for many households, though small businesses, construction work, and services like transportation and education also contribute to the local economy. While the literacy rate is relatively high compared to rural villages, there remains a gap in access to modern agricultural technologies, quality healthcare, and advanced market systems. Many small and marginal farmers depend on seasonal income, and face challenges such as rising input costs and unstable market prices.

However, the community shows a growing awareness of environmental sustainability and Chemical-free farming methods, which made our project on Organic Farming highly relevant and impactful. Despite limited resources, the people of Pileru are resilient, hardworking, and open to learning — key factors that made this project a mutually beneficial experience.



**Fig:** Grounding the Seeding

## CHAPTER 3: COMMUNITY SERVICE PART

### Description of the Activities Undertaken in the Community

As part of our Community Service Project, our team of five students from the Department of Computer Science and Engineering undertook a project focused on Organic Farming in and around Pileru, Annamayya District, Andhra Pradesh.

The objective of this project was to understand, participate in, and promote sustainable farming practices among local farmers and communities.

We carried out the following key activities during our engagement with the community:

**1. Field Visits and Observation:** We visited organic and traditional farms in the nearby villages around Pileru to observe their day-to-day farming practices. This helped us understand the difference between chemical-based and organic farming methods.

**2. Interactive Sessions with Farmers:** We held informal discussions with local farmers to learn about their traditional knowledge, the challenges they face, and their awareness of organic farming techniques.

**3. Participation in Organic Practices:** Our team actively participated in: Preparing organic compost using cow dung, kitchen waste, and dry leaves. Assisting in sowing seeds and natural pest control measures using neem-based solutions. Learning about crop rotation and mixed cropping practices.



**Fig:** Traditional Farming

**4. Awareness Session for Youth and Locals:** We conducted a small awareness session for school students and interested villagers on the benefits of organic farming, including health safety.

**5. Documentation and Feedback:** All observations and learning were documented. We also took feedback from farmers and villagers on how such student interactions could help them in the future.

### Reflections: Values, Life Skills, and Technical Skills Acquired

Participating in this project was a transformative experience. It allowed us to go beyond the classroom and interact with the real world.

Values Gained:

- ❖ Empathy and Social Responsibility: Understanding the challenges of farmers made us more empathetic and motivated to contribute positively to society.
- ❖ Life Skills Acquired:
- ❖ Communication and Teamwork: Interacting with community members improved our language and interpersonal skills.



# Organic Farming

## WHAT IS ORGANIC FARMING?

Organic farming is the use of agricultural production system reliant on green manure, compost, biological pest control, and crop rotation to produce crops, livestock and popularity. Organic centered agricultural production system fosters the cycling of the resources to conserve biodiversity and promote ecological balance. The use of green manure, cover crops, animal manure, and soil rotation , to interrupt the habitation of pests and diseases, improve soil fertility, and maximize the soil's biological activity.



Fig: Production of plats Through Organic Farming

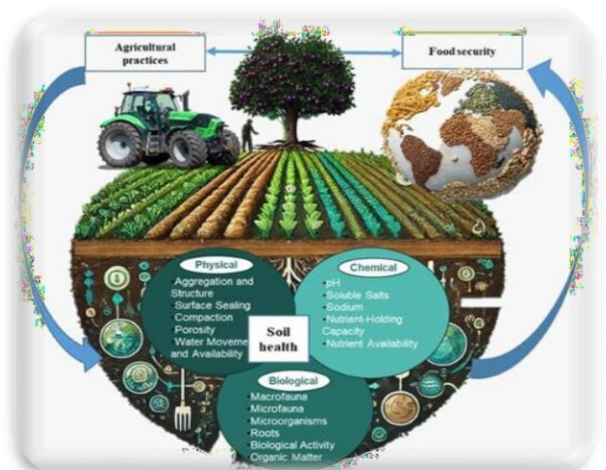
## OBJECTIVE OF ORGANIC FARMING:-

Organic farming may be adopted, to.....

1. Increase genetic diversity.
2. Promote more usage of natural pesticides.
3. Make sure the right soil cultivation at the right time.
4. Keep and build good soil structure and fertility.
5. Control pests, diseases and weed.

## TYPES OF ORGANIC FARMING:-

1. **Pure organic farming:-** It involves the use of organic manures and bio-pesticides with complete avoidance of inorganic chemicals and pesticides.
2. **Integrated organic farming:-** It involves integrated nutrients management and integrated pests management. It is the type of farming where you grow crops from natural resources. Further, having the complete nutritive value and also manage to prevent the crop or plants.



## METHOD OF ORGANIC FARMING:-

**1. Soil management:-** After cultivation of crops, the soil loses its nutrients and its quality deplete. Organic agriculture initiates the use of bacteria that is present in animal waste . The bacteria helps in making the soil nutrients more productive and fertile

**2. Weed management:-** Weed is unwanted plant that grows in agricultural fields. Organic agriculture focuses on lowering weed and not removing it completely. The two most widely used weed management techniques are-----

- Mulching: A process where we use plastic films or plant residue on the surface of the soil to block the growth of weed.
- Mowing or cutting:- Where there is a removal of weeds of top growth.



Fig: Methods Of Organic Farming

**3. Crop diversity:** Monoculture is the practice used in the agriculture fields where we harvest and cultivate only type of crop in a particular place. Recently , polyculture has come in existence . where we harvest and cultivate kinds of crops . To meet the increasing crop demand and produce the required soil microorganism.

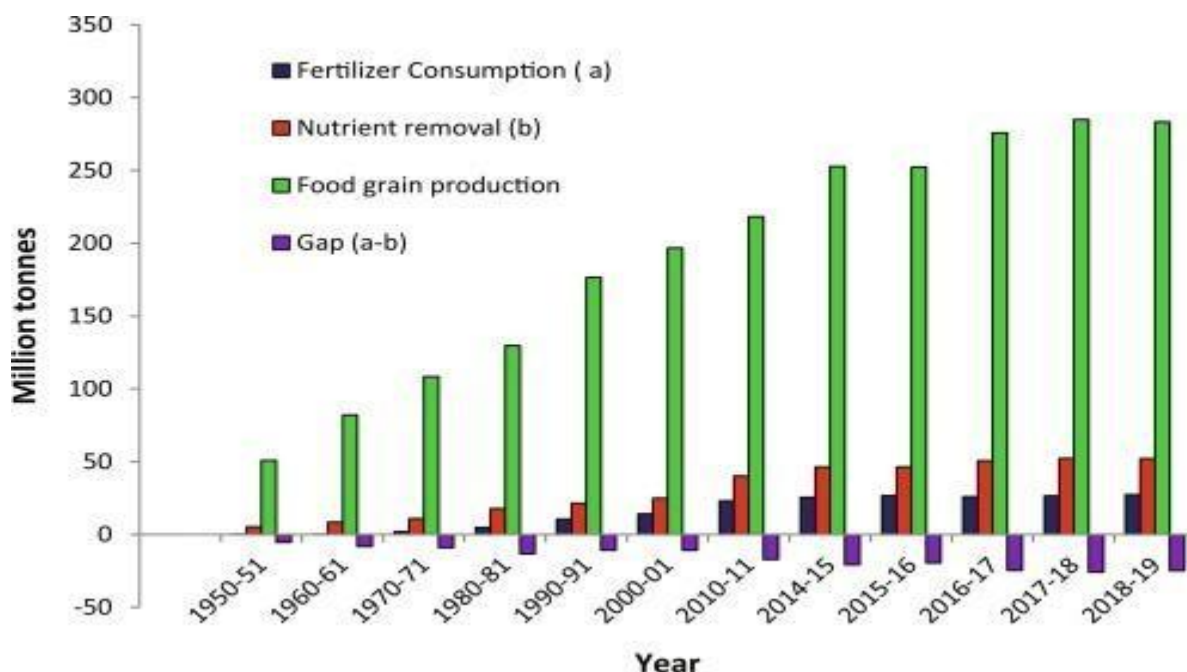
**4. Controlling other organism:-** There are both useful and harmful organism in the agricultural farm which affect the field , So, we need to control the growth of such organisms to protect the soil and the crops. We can do this by the use of herbicides and pesticides that contain fewer chemicals or are natural.



Fig: Growing Process of a Seed

## GROWTH OF ORGANIC FARMING:-

- ❖ In order to satisfy the increasing demand of organic products the organic farming sector in India has substantially increased over the count of year.
- ❖ India ranks 33 in world in terms of area under organic farming .
- ❖ India rank 88" in terms of the ratio of agriculture land under organic crop to total farming area.
- ❖ Madhya Pradesh has highest area under organic farming (1 m or 52%) .
- ❖ Maharashtra is at second ( 0 mho33.6%) .
- ❖ Orissa is third (0.67mhar or 9.7%).



**Fig:** A prospect for food, environment and livelihood security in Indian agriculture

## Sikkim Model:

- ❖ On 9 January 2016 Honorable prime minister declared Sikkim as First Organic State in India.
- ❖ Sikkim win the Future Policy Award, a prize given to the best global policies in the field of agroecology, organised by the FAO together with the World Future Council and IFOAM Organics International.
- ❖ The State was awarded with a citation from Agriculture Today in the year 2009 for 'Outstanding Performance in Horticulture.
- ❖ Sikkim received Development received UN word for becoming world's first Organic State, Beats 25 countries.

### ACTIVITY LOG FOR THE FIRST WEEK

| <b>DAY<br/>&amp;<br/>DATE</b> | <b>BRIEF DESCRIPTION OF THE DAILY<br/>ACTIVITY</b> | <b>LEARNING OUTCOME</b> | <b>Person<br/>In-charge<br/>Signature</b> |
|-------------------------------|----------------------------------------------------|-------------------------|-------------------------------------------|
| <b>Day -<br/>1</b>            |                                                    |                         |                                           |
| <b>Day -<br/>2</b>            |                                                    |                         |                                           |
| <b>Day -<br/>3</b>            |                                                    |                         |                                           |
| <b>Day -<br/>4</b>            |                                                    |                         |                                           |
| <b>Day -<br/>5</b>            |                                                    |                         |                                           |
| <b>Day -<br/>6</b>            |                                                    |                         |                                           |



## WEEKLY REPORT

**WEEK – 1 (From Dt..... to Dt ..... )**

[illegible]

**ACTIVITY LOG FOR THE SECOND WEEK**

| <b>DAY<br/>&amp;<br/>DATE</b> | <b>BRIEF DESCRIPTION OF THE<br/>DAILY ACTIVITY</b> | <b>LEARNING<br/>OUTCOME</b> | <b>Person<br/>In-charge<br/>Signature</b> |
|-------------------------------|----------------------------------------------------|-----------------------------|-------------------------------------------|
| <b>Day –<br/>1</b>            |                                                    |                             |                                           |
| <b>Day-<br/>2</b>             |                                                    |                             |                                           |
| <b>Day –<br/>3</b>            |                                                    |                             |                                           |
| <b>Day –<br/>4</b>            |                                                    |                             |                                           |
| <b>Day –<br/>5</b>            |                                                    |                             |                                           |
| <b>Day –<br/>6</b>            |                                                    |                             |                                           |

## WEEKLY REPORT

**WEEK – 2 (From Dt..... to Dt..... )**

[illegible]

**ACTIVITY LOG FOR THE THIRD WEEK**

| <b>DAY<br/>&amp;<br/>DATE</b> | <b>BRIEF DESCRIPTION OF THE DAILY<br/>ACTIVITY</b> | <b>LEARNING OUTCOME</b> | <b>Person<br/>In-charge<br/>Signature</b> |
|-------------------------------|----------------------------------------------------|-------------------------|-------------------------------------------|
| <b>Day -<br/>1</b>            |                                                    |                         |                                           |
| <b>Day-<br/>2</b>             |                                                    |                         |                                           |
| <b>Day -<br/>3</b>            |                                                    |                         |                                           |
| <b>Day -<br/>4</b>            |                                                    |                         |                                           |
| <b>Day -<br/>5</b>            |                                                    |                         |                                           |
| <b>Day -<br/>6</b>            |                                                    |                         |                                           |

## WEEKLY REPORT

**WEEK – 3 (From Dt..... to Dt..... )**

[illegible]

**ACTIVITY LOG FOR THE FOURTH WEEK**

| <b>DAY<br/>&amp;<br/>DATE</b> | <b>BRIEF DESCRIPTION OF THE DAILY<br/>ACTIVITY</b> | <b>LEARNING OUTCOME</b> | <b>Person<br/>In-charge<br/>Signature</b> |
|-------------------------------|----------------------------------------------------|-------------------------|-------------------------------------------|
| <b>Day -<br/>1</b>            |                                                    |                         |                                           |
| <b>Day-<br/>2</b>             |                                                    |                         |                                           |
| <b>Day -<br/>3</b>            |                                                    |                         |                                           |
| <b>Day -<br/>4</b>            |                                                    |                         |                                           |
| <b>Day -<br/>5</b>            |                                                    |                         |                                           |
| <b>Day -<br/>6</b>            |                                                    |                         |                                           |

## WEEKLY REPORT

**WEEK - 4 (From Dt..... to Dt.....)**

[illegible]



### ACTIVITY LOG FOR THE FIFTH WEEK

| <b>DAY<br/>&amp;<br/>DATE</b> | <b>BRIEF DESCRIPTION OF THE DAILY<br/>ACTIVITY</b> | <b>LEARNING OUTCOME</b> | <b>Person<br/>In-charge<br/>Signature</b> |
|-------------------------------|----------------------------------------------------|-------------------------|-------------------------------------------|
| <b>Day -<br/>1</b>            |                                                    |                         |                                           |
| <b>Day-<br/>2</b>             |                                                    |                         |                                           |
| <b>Day -<br/>3</b>            |                                                    |                         |                                           |
| <b>Day -<br/>4</b>            |                                                    |                         |                                           |
| <b>Day -<br/>5</b>            |                                                    |                         |                                           |
| <b>Day -<br/>6</b>            |                                                    |                         |                                           |

## WEEKLY REPORT

**WEEK – 5 (From Dt..... to Dt..... )**

[illegible]

## CHAPTER 5: OUTCOMES DESCRIPTION

### Socio-Economic Survey – Pileru (Annamayya District)

As part of our Community Service Project, we conducted a Socio-Economic Survey in the village of Pileru, located in Annamayya district. The purpose of this survey was to understand the lifestyle, income sources, education level, farming practices, and the challenges faced by the local residents — especially organic farmers.

We visited several households and interacted with farmers, homemakers, youth, and elders. The findings helped us understand the socio-economic background of the community and tailor our organic farming awareness activities accordingly.

#### Summary of Survey Findings:

- Main Occupation: Agriculture and daily labor
- Average Family Income: ₹5,000 – ₹12,000 per month
- Education Level: Many adults have primary education; youth are pursuing higher education in nearby towns
- Farming Practice: Traditional and organic methods are followed by some; others still rely on chemical farming Common
- Problems: Low crop yield, water scarcity, lack of market support of organic products Scheme
- Awareness: Some farmers are aware of government subsidies and organic farming program
- Healthcare & Facilities: Basic healthcare and education facilities are available, but limited.



**Fig:** Observing the field of crops

## Problems Identified in the Community–Pileru (Annamayya District)

During our Community Service Project on Organic Farming, we interacted with farmers and local residents in Pileru and identified the following keys of problems faced by the community:

1. **Lack of Awareness about Organic Farming:** Many farmers were unaware of the long-term benefits of organic farming and continued using chemical fertilizer and pesticides due to habit or lack of training.
2. **Limited Access to Organic Inputs:** Farmers reported difficulty in accessing natural fertilizers, bio-pesticides, and compost materials. This discouraged them from shifting to organic practices.
3. **Low Market Support for Organic Produce:** Even when farmers tried organic methods, they faced challenges in selling their produce at good prices due to lack of separate organic markets.
4. **Financial Constraints:** Many small farmers could not afford the initial cost of transitioning to organic farming or lacked support from banks and agricultural loans.



**Fig:** Healthy food through Organic Farming Only

5. **Water Scarcity:** Irregular rainfall and lack of irrigation infrastructure affected crop growth, especially for farmers using traditional and organic methods.
6. **Lack of Training and Guidance:** There is a shortage of workshops, field visits, or training programs to guide farmers in adopting modern organic techniques.
7. **Youth Migration:** Many young people are moving to cities for jobs, leading to a shortage of labor and fewer people interested in continuing farming.



## **Short-Term Action Plan**

1. **Conduct Awareness Campaigns:** Organize regular awareness sessions and demonstrations on the benefits of organic farming for local farmers and youth.
2. **Distribute Organic Farming Kits:** Provide compost samples, neem-based bio-pesticides, and basic natural farming materials through NGOs or government schemes.
3. **Encourage Farmer Groups:** Motivate small and marginal farmers to form Organic Farmer Groups to share knowledge, inputs, and resources.
4. **Involve Students and Youth Volunteers:** Promote student-led farming assistance programs during weekends or holidays to reduce labor shortages.

## **Long-Term Action Plan**

1. **Set Up a Local Organic Market:** Establish a separate space for the sale of organic produce to attract better prices and customer trust.
2. **Irrigation System Improvement:** Recommend installation of low-cost drip irrigation systems with government support to solve water scarcity issues.
3. **Training and Certification Programs:** Recommend to the Agriculture department to conduct regular certified training sessions on organic methods for farmers.
4. **Financial Support and Loans:** Suggest to local banks and authorities to offer low-interest loans or subsidies specifically for organic farmers.

## Community Awareness Programme – Description & Outcomes

As part of our Community Service Project on organic farming, we conducted awareness programmes in Kadirepalli village to address the key problems identified, especially the lack of awareness and limited adoption of organic practices.

### ❖ Programme Activities Conducted:

1. **Awareness Session at Local School:** We interacted with school students and explained the importance of organic farming through simple language, posters, and real-life examples. We also encouraged them to grow small plants at home.
2. **Farmer Interaction & Poster Campaign:**  
We visited farms and distributed hand-made posters that highlighted:
  - i. Benefits of organic farming
  - ii. Differences between chemical and organic methods
  - iii. How to make simple compost at home
3. **Live Composting Demonstration:** We
4. conducted a live demo on preparing compost using vegetable waste, dry leaves, and cow dung to show how easy and cost-effective organic inputs can be.
5. **Door-to-Door Visits:** Our team visited several households and had one-on-one discussions about farming practices, sharing simple tips and answering questions about natural farming.



**Fig:** Farmers meeting on the field

### ❖ Outcomes of the Programme:

- > Many farmers showed interest in trying organic methods after seeing the compost demo.
- > Students and youth became more aware and curious about farming and environmental issues.



# Project Report

## Awareness and Field Study on Organic Farming

### Introduction:

As part of our Community Service Project, we conducted a mini-project focused on organic farming practices in the village of Pileru, Annamayya district. The aim of this mini-project was to understand the local farming methods and create awareness among farmers about the benefits of organic farming.

Organic farming has engrossed much attention in current decades as a way to maintain farming production. At the same time, it has played an important role in dealing with the environmental harms rooted in traditional agricultural techniques.



Fig: Showing different type of Organic Farming.

Organic farming not only produces fine and healthy food products but also improves the fertility and quality of soil (Isaacs 2012). Organic agriculture requires low external inputs, thereby contributing to increased food availability. It is a system based on generating food with negligible destruction to ecosystems, humans, or animals.

Weed management in organic farming is not a simple job, mainly in regions where labor for hand weeding is not reasonably priced or is scarce. On the other hand, the standard must be similar as in any traditional cropping scheme, i.e., weed competition desires to be prohibited in turn to attain the highest crop yields. This essentially involves weeding with nonchemical substances, however, this has to be done accurately at the precise time to get rid of weeds during the supposed significant phase of weed competition. Organic schemes also need the use of precautionary techniques prior to raising the crop and to set up a sensible crop rotation. Decayed seedbed preparation to kill the weeds manually or mechanically is an extremely fine choice to hold up the launch of weed competition.

## 1. Objectives of the Mini Project:

- To observe and understand the farming practices used by local farmers
- To spread awareness about the importance and benefits of organic farming
- To interact with the farming community and understand their problems
- To share simple organic farming techniques with farmers
- To promote the use of natural fertilizers and pesticides



**Fig:** observe and understand the farming practices used by local farmers

Organic farming is an agricultural system that uses fertilizers of Organic origin such as compost manure, green manure, and bone meal and place emphasis on techniques such as crop rotation and companion planting. It originated early in the 20th century in reaction to rapidly changing farming practices.

### Crop Rotation:

Crop rotations are at the heart of organic agriculture and aid organic methods to shield our surrounding. They engage in altering the form of crop developed in one area on a usual basis. Organic farmers sow alternating groups of plants (brassica, cereals, roots, and legumes) to put in fertility and stop diseases and pests from strengthening.



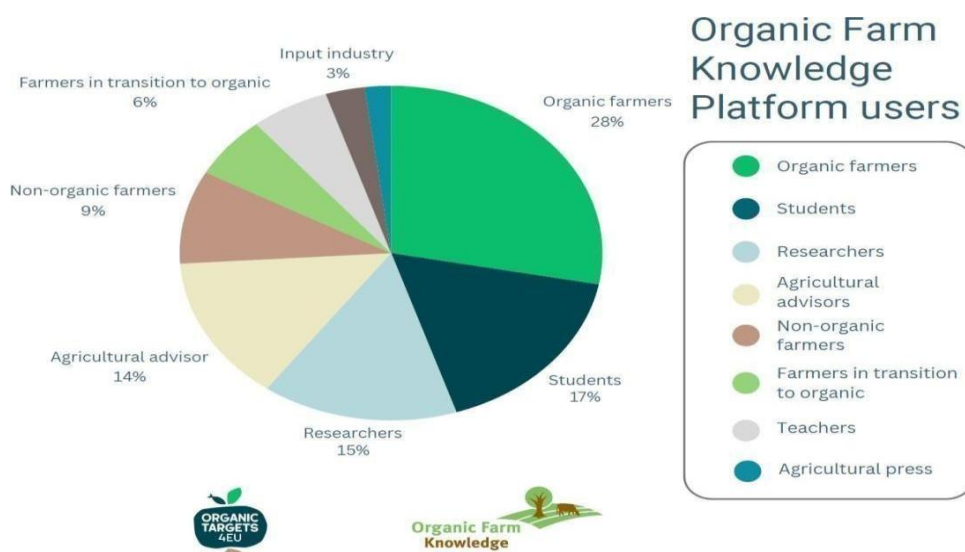
## 2. Methodology:

We visited different farms in the village and spoke directly with the farmers. We collected information through face-to-face conversations and used printed forms to note down their responses. We also explained organic farming concepts using posters and simple materials prepared by our team.



Fig: visited different farms in the village and observed

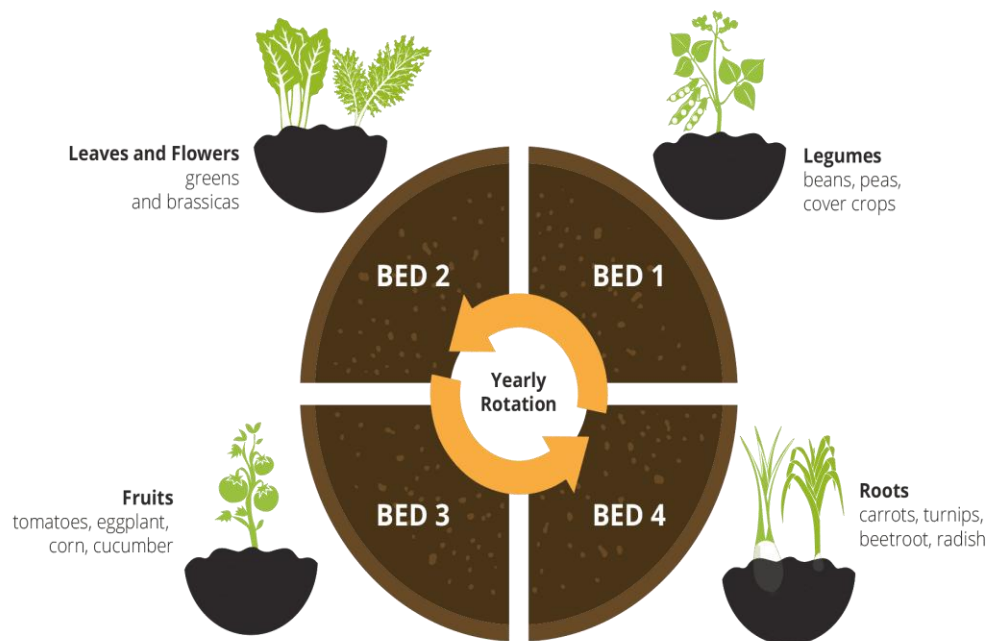
Organic farming may have an important social effect on local communities. To start with, organic agriculture may provide employment opportunities to the local people. More manual labor is often required in organic agriculture fertilizer loss.



### 3. Observations and Findings:

During our visits, we observed that:

- ✓ Many farmers still rely on chemical fertilizers and pesticides.
- ✓ A few farmers had tried composting but lacked proper knowledge.
- ✓ Interest in organic farming was present, but awareness was low.
- ✓ Farmers were open to learning and appreciated our efforts.



**Fig:** Different types of Crop Rotation

Although using crop rotation may sound old fashioned, it is a more effective and sophisticated method than chemicals. Using chemicals to fertilize the soil frequently merely supplies crops with three basic elements (nitrogen, potassium, and phosphorous) that they require to grow before supplying them with each and every nutrient that they require. When a farmer grows the similar crop in one field year after year (known as monoculture), the pests and diseases that attack the crop set up and get enlarged in number with time.

#### 4. Awareness and Suggestions :

We conducted awareness programs in small groups and explained the benefits of organic farming. We encouraged farmers to try composting, avoid overuse of chemicals, and use natural pest control methods. We suggested seeking help from agricultural officers for guidance and attending workshops organized by the government or NGOs.

c



**Fig:** Using natural pest control methods

Food for starving population, fiber for clothing, and feed for animals and even, in a number of cases, fuel for vehicles come from worldwide agriculture. Consequently, in the world's temperate climates, human agriculture has displaced 45 % of temperate forests, 50 % of savannas, and 70% of grasslands. Agriculture is one of the main sources of GHG emissions; the most important cause of deforestation in the tropics and a recurrent basis of water pollution and nonrenewable groundwater mining.

A number of farmers have turned to the organic methods. Such a kind of farming is destined to reduce human health and environmental impacts by evading the use of chemical pesticides, synthetic fertilizers, and antibiotic or hormone treatment for livestock. The use of industrial methods, predominantly synthetic nitrogen fertilizer, has fed the human population during the previous century.



## Chemical Control:

The pest problem cannot be solved by the pesticides. Insecticide use of that has amplified ten fold , whereas crop fatalities from pest spoil have doubled in that past 50 years.

Three significant causes of why natural power is preferable to pesticide use are given below:

### ❖ Safety For People:

Synthetic pesticides can rapidly discover their mean into watercourses and food chains. This can generate health risks for humans. Human health can also be debilitated by population consuming food (especially vegetables and fruit) holding remains of pesticides sprayed on the produce. Around the world, there are an estimated 1 million cases of poisoning by pesticides each year. Approximately 20,000 of these results in death. The majority of the deaths occur in tropical countries, where chemical pesticides prohibited in the USA and Europe are still available.



### ❖ Cost:

- As Natural techniques do not involve purchasing matter from the exterior, using natural disease and pest control is frequently cheaper than applying chemical Pesticides.



### ❖ Avoiding:

Chemical pesticides can have a number of harmful effects on the environment. They are:

- Useful insects can be killed by chemical pesticides. The equilibrium between pests and helpful predators can be disturbed by only one spray.
- Artificial chemicals can reside in the bodies of animals and in the environment causing troubles for several years.

## Recommendations and Conclusion

### Recommendations:

Based on our observations and interactions with the farmers in Pileru village, we suggest the following steps to encourage and support organic farming:

1. Conduct regular awareness sessions in the village to educate farmers about the benefits of organic farming.
2. Distribute pamphlets or posters in local language showing simple methods for composting and natural pesticide preparation.
3. Encourage the use of cow dung, compost, and neem-based sprays as natural alternatives to chemical inputs.
4. Involve agricultural officers or organic farming experts to visit the village and guide farmers practically.
5. Motivate youth in the village to learn about organic farming and help spread the knowledge to their families.

### Conclusions:

Through this community service project, we learned that:

- Many farmers are interested in organic farming but lack guidance and confidence.
- Most of them still use chemical fertilizers and pesticides because they are unaware of natural alternatives.
- Simple awareness activities by students can create a positive change in mindset.
- Organic farming not only protects the soil and environment but also improves the health of consumers.

This project helped us develop communication skills, understand rural life closely, and inspired us to use our knowledge for community development.

## Student Self-Evaluation for the Community Service Project

Student Name:

Registration No:

Period of CSP: From: To:

Date of Evaluation:

Name of the Person in-charge:

**Please rate your performance in the following areas:**

**Rating Scale:** 1 is lowest and 5 is highest rank

|                                       |   |   |   |   |   |
|---------------------------------------|---|---|---|---|---|
| 1) Oral communication                 | 1 | 2 | 3 | 4 |   |
| 2) Written communication              | 1 | 2 | 3 | 4 | 5 |
| 3) Proactiveness                      | 1 | 2 | 3 | 4 | 5 |
| 4) Interaction ability with community | 1 | 2 | 3 | 4 | 5 |
| 5) Positive Attitude                  | 1 | 2 | 3 | 4 | 5 |
| 6) Self-confidence                    | 1 | 2 | 3 | 4 | 5 |
| 7) Ability to learn                   | 1 | 2 | 3 | 4 | 5 |
| 8) Work Plan and organization         | 1 | 2 | 3 | 4 | 5 |
| 9) Professionalism                    | 1 | 2 | 3 | 4 | 5 |
| 10) Creativity                        | 1 | 2 | 3 | 4 | 5 |
| 11) Quality of work done              | 1 | 2 | 3 | 4 | 5 |
| 12) Time Management                   | 1 | 2 | 3 | 4 | 5 |
| 13) Understanding the Community       | 1 | 2 | 3 | 4 | 5 |
| 14) Achievement of Desired Outcomes   | 1 | 2 | 3 | 4 | 5 |
| 15) OVERALL PERFORMANCE               | 1 | 2 | 3 | 4 |   |

Date:

Signature of the Student

## Evaluation by the Person in-charge in the Community/Habitation

Student Name:

Registration No:

Period of CSP: From: To:

Date of Evaluation:

Name of the Person in-charge:

Please rate the student's performance in the following areas:

Please note that your evaluation shall be done independent of the Student's self-evaluation

Rating Scale: 1 is lowest and 5 is highest rank

|                                       |   |   |   |   |   |
|---------------------------------------|---|---|---|---|---|
| 1) Oral communication                 | 1 | 2 | 3 | 4 |   |
| 2) Written communication              | 1 | 2 | 3 | 4 | 5 |
| 3) Proactiveness                      | 1 | 2 | 3 | 4 |   |
| 4) Interaction ability with community | 1 | 2 | 3 | 4 | 5 |
| 5) Positive Attitude                  | 1 | 2 | 3 | 4 | 5 |
| 6) Self-confidence                    | 1 | 2 | 3 | 4 | 5 |
| 7) Ability to learn                   | 1 | 2 | 3 | 4 |   |
| 8) Work Plan and organization         | 1 | 2 | 3 | 4 | 5 |
| 9) Professionalism                    | 1 | 2 | 3 | 4 | 5 |
| 10) Creativity                        | 1 | 2 | 3 | 4 | 5 |
| 11) Quality of work done              | 1 | 2 | 3 | 4 |   |
| 12) Time Management                   | 1 | 2 | 3 | 4 | 5 |
| 13) Understanding the Community       | 1 | 2 | 3 | 4 | 5 |
| 14) Achievement of Desired Outcomes   | 1 | 2 | 3 | 4 | 5 |
| 15) OVERALL PERFORMANCE               | 1 | 2 | 3 | 4 | 5 |

Date:

Signature of the Supervisor



## PHOTOS AND VIDEO LINKS

